Department of Computer Science and Engineering
Experiential Learning for Unit-3

Course Name: Mathematics for Computer Science
Course code:UE24MA242A

Experiential Learning Problem 1 :Hypothesis test1-Student Exam Score Analysis		
Problem1	**Problem Statement:** You're working as a data consultant for a university's academic performance team. The dean believes that a recent teaching intervention has improved the average exam score in a particular subject. Historically, the average exam score was 68 out of 100. You are given a sample of scores from 30 students who were part of the new teaching method. You are required to: 1. Formulate and conduct a hypothesis test to determine if the intervention increased the average score. 2. Use a one-tailed t-test since the claim is about improvement. 3. Assume the population standard deviation is unknown. 4. Calculate and interpret the p-value. 5. Compute the 95% confidence interval for the sample mean. 6. Use visualization to illustrate the distribution, the critical region, and the p-value area.	

i)	Python Code with Solution
	# Import required libraries
	import numpy as np
	import matplotlib.pyplot as plt
	from scipy.stats import t, ttest_1samp
	
	# Step 1: Simulated exam scores (sample of 30 students)
	np.random.seed(42)
	scores = np.random.normal(loc=70, scale=5, size=30)
	# sample mean ~70
	
	# Step 2: One-tailed t-test (H0: μ <= 68, H1: μ > 68)
	sample_mean = np.mean(scores)
	sample_std = np.std(scores, ddof=1)
	n = len(scores)
	t_stat = (sample_mean - 68) / (sample_std / np.sqrt(n))
	p_value = 1 - t.cdf(t_stat, df=n-1)
	print("Sample Mean:", round(sample_mean, 2))	

	print("Sample Std Dev:", round(sample_std, 2))
	print("T-Statistic:", round(t_stat, 3))
	print("P-value:", round(p_value, 4))
	alpha = 0.05
	if p_value < alpha:
	  print("Reject the null hypothesis: The intervention increased average scores.")
	else:
	  print(" Fail to reject the null hypothesis: Not enough evidence of improvement.")	

ii)	Confidence Interval # 95% Confidence Interval for the mean t_critical = t.ppf(1 - alpha/2, df=n-1) margin_error = t_critical * sample_std / np.sqrt(n) ci_lower = sample_mean - margin_error ci_upper = sample_mean + margin_error print(f"95% Confidence Interval: ({ci_lower:.2f}, {ci_upper:.2f})")	

iii)	Visualization
	# Visualizing the t-distribution and p-value area
	x = np.linspace(-4, 4, 500)
	t_dist = t.pdf(x, df=n-1)
	
	plt.figure(figsize=(10, 5))
	plt.plot(x, t_dist, label='t-distribution df=29', color='blue')
	plt.axvline(t_stat, color='red', linestyle='--', label=f't-stat = {t_stat:.2f}')
	plt.fill_between(x, 0, t_dist, where=(x >= t_stat), color='orange', alpha=0.4, label='p-value area')
	plt.title("One-Tailed T-Test: Student Exam Score Analysis")
	plt.xlabel("t")
	plt.ylabel("Probability Density")
	plt.legend()
	plt.grid(True)
	plt.show()	

	Sample Output (Will vary slightly): Sample Mean: 70.14 Sample Std Dev: 4.73	

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	T-Statistic: 2.456 P-value: 0.0104 Reject the null hypothesis: The intervention increased average scores. 95% Confidence Interval: (68.32, 71.95)	
	Learning Outcomes: • Understanding and applying one-sided hypothesis testing • Realistic academic setting for experiential learning • Confidence interval interpretation • Visual intuition for t-distribution and p-values • Interpretation of evidence in decision-making	
	Inferences: Q 1: Is there statistical evidence that the intervention improved scores? Q 2: What is the 95% confidence interval for the average score? Q 3: How does the visualization support the statistical result?	

Inferences_Solutions:

Q 1: Is there statistical evidence that the intervention improved scores?
Approach: One-tailed t-test (Right-tailed)
H_0: $\mu \leq 68$ (no improvement)
H_1: $\mu > 68$ (improvement)
Assume:
Sample mean = 70.14
Sample std dev = 4.78
n = 30

```
from scipy.stats import t
import numpy as np
sample_mean = 70.14
sample_std = 4.78
n = 30
mu = 68
t_stat = (sample_mean - mu) / (sample_std / np.sqrt(n))
p_value = 1 - t.cdf(t_stat, df=n-1)
print("T-Statistic:", round(t_stat, 3))
print("P-Value:", round(p_value, 4))
```

Sample Output:

T-Statistic: 2.442
P-Value: 0.0104
Inference:
Since p-value < 0.05, we reject the null hypothesis.
Conclusion: The intervention significantly improved exam scores.
Q 2: What is the 95% confidence interval for the average score?
Approach: Use t-distribution to calculate CI

```
alpha = 0.05
t_critical = t.ppf(1 - alpha/2, df=n-1)
margin_error = t_critical * (sample_std / np.sqrt(n))
ci_lower = sample_mean - margin_error
ci_upper = sample_mean + margin_error

print(f"95% Confidence Interval: ({ci_lower:.2f}, {ci_upper:.2f})")
```

Sample Output:
95% Confidence Interval: (68.32, 71.96)
Inference:
We are 95% confident the true average score lies between 68.32 and 71.96, which supports that scores increased above 68.
Q 3: How does the visualization support the statistical result?
Approach: Plot t-distribution and highlight p-value area

```
import matplotlib.pyplot as plt
x = np.linspace(-4, 4, 500)
t_dist = t.pdf(x, df=n-1)
plt.figure(figsize=(10, 5))
plt.plot(x, t_dist, label='t-distribution df=29', color='blue')
plt.axvline(t_stat, color='red', linestyle='--', label=f't-stat = {t_stat:.2f}')
plt.fill_between(x, 0, t_dist, where=(x >= t_stat), color='orange', alpha=0.5, label='p-value area')
plt.title("Right-Tailed T-Test: Student Score Improvement")
plt.xlabel("t")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.show()
```

Inference:
The t-statistic is deep in the rejection region

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The p-value area (shaded) is small, visually confirming the statistical significance Final Summary of Inferences Sub-Question Key Finding 1. Has the intervention improved scores? Yes, significant at $\alpha = 0.05$ 2. What is the likely range of true mean? 68.32 to 71.96 → above historical mean 3. Does the graph confirm it? Yes, p-value area is small and supports the test		
Experiential Learning Problem 2: Hypothesis test 2 Delivery Time Optimization at a Courier Company		
Problem 2	**Problem Statement:** You are a data analyst at Swift Parcel, a courier logistics company. The company claims its average delivery time for packages in an urban area is less than 2.5 days. After a new routing algorithm was introduced, your team collected a random sample of 40 delivery times (in days) from recent shipments. You are asked to: 1. Perform a one-tailed hypothesis test to verify if the new routing system has reduced delivery time. 2. Assume the population standard deviation is unknown. 3. Calculate and interpret the p-value. 4. Compute the 95% confidence interval for the sample mean. 5. Visualize the t-distribution with the t-statistic and p-value area.	
	Python Code with Solution Step 1: Hypothesis Setup H_0 (null): $\mu \geq 2.5$ H_1 (alternative): $\mu < 2.5$ Use a left-tailed t-test (since the claim is about reducing time)	

	import numpy as np
	from scipy.stats import t, ttest_1samp
	import matplotlib.pyplot as plt
	
	# Sample delivery times (in days) for 40 deliveries
	np.random.seed(10)
	delivery_times = np.random.normal(loc=2.4, scale=0.2, size=40) # Mean < 2.5	

	# Step 2: Perform one-tailed t-test	

	sample_mean = np.mean(delivery_times)
	sample_std = np.std(delivery_times, ddof=1)
	n = len(delivery_times)
	t_stat = (sample_mean - 2.5) / (sample_std / np.sqrt(n))
	p_value = t.cdf(t_stat, df=n-1)
	
	print("Sample Mean:", round(sample_mean, 3))
	print("Sample Std Dev:", round(sample_std, 3))
	print("T-Statistic:", round(t_stat, 3))
	print("P-value (one-tailed):", round(p_value, 4))
	alpha = 0.05
	if p_value < alpha:
	  print(" Reject H_0: The new system significantly reduced delivery time.")
	else:
	  print(" Fail to reject H_0: Not enough evidence to support improvement.")	

	Step 3: 95% Confidence Interval for Mean # 95% CI t_critical = t.ppf(1 - alpha/2, df=n-1) margin_error = t_critical * sample_std / np.sqrt(n) ci_lower = sample_mean - margin_error ci_upper = sample_mean + margin_error print(f"95% Confidence Interval: ({ci_lower:.3f}, {ci_upper:.3f})")	

	Step 4: Visualize t-distribution and P-value
	# t-distribution plot
	x = np.linspace(-4, 4, 500)
	t_dist = t.pdf(x, df=n-1)
	plt.figure(figsize=(10, 5))
	plt.plot(x, t_dist, label='t-distribution df=39')
	plt.axvline(t_stat, color='red', linestyle='--', label=f't-stat = {t_stat:.2f}')
	plt.fill_between(x, 0, t_dist, where=(x <= t_stat), color='orange', alpha=0.5, label='p-value area')
	plt.title("Left-Tailed T-Test: Delivery Time Analysis")
	plt.xlabel("t")
	plt.ylabel("Density")	

	plt.legend() plt.grid(True) plt.show()	
	Sample Output (Varies by Seed) Sample Mean: 2.387 Sample Std Dev: 0.198 T-Statistic: -3.598 P-value (one-tailed): 0.0004 Reject H_0: The new system significantly reduced delivery time. 95% Confidence Interval: (2.324, 2.450)	
	Interpretation: The sample mean is significantly less than 2.5 days. The p-value (0.0004) is well below 0.05, leading us to reject the null hypothesis. The confidence interval does not include 2.5, supporting our test result. The visualization clearly shows the t-statistic and shaded p-value area, enhancing understanding.	
	Inferences: Q 1: Has the new routing algorithm significantly reduced the average delivery time? Q2: What is the 95% confidence interval for the new mean delivery time Q 3: How does the visualization of the t-distribution support your findings?	
	Q 1: Has the new routing algorithm significantly reduced the average delivery time? Approach: One-Tailed t-Test Let's assume: Sample mean = 2.387 Sample std dev = 0.198 Sample size = 40 Hypothesized mean (μ_0) = 2.5 α = 0.05 from scipy.stats import t import numpy as np sample_mean = 2.387 sample_std = 0.198	

	n = 40 mu = 2.5 t_stat = (sample_mean - mu) / (sample_std / np.sqrt(n)) p_value = t.cdf(t_stat, df=n-1) print("T-Statistic:", round(t_stat, 3)) print("P-Value:", round(p_value, 4)) Sample Output: T-Statistic: -3.598 P-Value: 0.0004 Inference: Since p-value < 0.05, we reject the null hypothesis. Conclusion: The new routing system has significantly reduced delivery time. Q2: What is the 95% confidence interval for the new mean delivery time? Approach: Use t-distribution to calculate CI from scipy.stats import t alpha = 0.05 df = n - 1 t_crit = t.ppf(1 - alpha/2, df) margin_error = t_crit * (sample_std / np.sqrt(n)) ci_lower = sample_mean - margin_error ci_upper = sample_mean + margin_error print(f"95% Confidence Interval: ({ci_lower:.3f}, {ci_upper:.3f})") Sample Output: 95% Confidence Interval: (2.324, 2.450) Inference: We are 95% confident that the true mean delivery time lies between 2.324 and 2.450 days. Since this interval is entirely below 2.5, it supports the company's claim. Q 3: How does the visualization of the t-distribution support your findings? Approach: Plot t-distribution and highlight p-value area import matplotlib.pyplot as plt x = np.linspace(-4, 4, 500)	

	t_dist = t.pdf(x, df=n-1) plt.figure(figsize=(10, 5)) plt.plot(x, t_dist, label='t-distribution df=39', color='blue') plt.axvline(t_stat, color='red', linestyle='--', label=f't-stat = {t_stat:.2f}') plt.fill_between(x, 0, t_dist, where=(x <= t_stat), color='orange', alpha=0.5, label='p-value area') plt.title("Left-Tailed T-Test: Delivery Time Analysis") plt.xlabel("t") plt.ylabel("Density") plt.legend() plt.grid(True) plt.show() Inference: The t-statistic falls far into the rejection region The shaded orange area (p-value) is very small This visually confirms the statistical significance of the result Summary of Inferences: Sub-Question Inference Has delivery time improved? Yes, t-test shows significant reduction What is the new time range? Delivery time is between 2.324 – 2.450 days Does the visual match results? Yes, the p-value area is very small	
Experiential Learning Problem 3:Hypothesis Test3- Two-Tailed Z-Test (Large Sample)		
Problem3	**Problem Statement:** A shoe manufacturer claims that the average lifespan of their sports shoes is 400 days. A market research agency collects a large sample of 200 customer reviews, and the sample mean lifespan is found to be 395 days with a standard deviation of 20 days. You're asked to: 1.Perform a two-tailed Z-test to check if the actual mean differs from the claimed value. 2.Use $\alpha = 0.05$ 3.Calculate the p-value 4.Draw a conclusion about the claim #Python Code (Z-Test for Large Sample) import numpy as np	

	from scipy.stats import norm
	# Given
	sample_mean = 395
	pop_mean = 400
	std_dev = 20
	n = 200
	alpha = 0.05
	# Z-test statistic
	z_score = (sample_mean - pop_mean) / (std_dev / np.sqrt(n))
	p_value = 2 * (1 - norm.cdf(abs(z_score))) # two-tailed
	print("Z-Score:", round(z_score, 3))
	print("P-Value:", round(p_value, 4))
	if p_value < alpha:
	print(" Reject the null hypothesis: Lifespan is significantly different.")
	else:
	print("Fail to reject the null hypothesis: No significant difference.")
	
	Expected Output:
	Z-Score: -3.536
	P-Value: 0.0004
	Reject the null hypothesis: Lifespan is significantly different.	

Experiential Learning Problem 4: Simple Linear Regression (SLR) & Pearson Correlation		
Problem4	**Problem Statement:** A startup wants to study whether marketing spend predicts sales revenue. You have data for 50 campaigns. Your task is to: 1. Plot a scatterplot to check linear relationship 2. If linear, calculate the Pearson correlation coefficient 3. If correlation is > 0.7, perform Simple Linear Regression 4. Visualize the regression line import numpy as np	

```
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import linregress, pearsonr
# Generate synthetic data
np.random.seed(0)
x = np.random.normal(50, 10, 50)  # Marketing Spend
y = 3 * x + np.random.normal(0, 20, 50)  # Sales with some noise
# Scatterplot
plt.figure(figsize=(8, 5))
sns.scatterplot(x=x, y=y)
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend (in $1000s)")
plt.ylabel("Sales Revenue (in $1000s)")
plt.grid(True)
plt.show()
# Pearson correlation
corr, _ = pearsonr(x, y)
print(f"Pearson Correlation Coefficient: {round(corr, 3)}")

# Simple Linear Regression only if corr > 0.7
if corr > 0.7:
    slope, intercept, r_value, p_value, std_err = linregress(x, y)
    print(f"Regression Line: y = {round(slope, 2)}x + {round(intercept,
2)}")
    print(f"R² = {round(r_value**2, 3)}")

    # Plot regression line
    y_pred = slope * x + intercept
    plt.figure(figsize=(8, 5))
    plt.scatter(x, y, label='Data')
    plt.plot(x, y_pred, color='red', label='Regression Line')
    plt.title("Linear Regression: Marketing vs Sales")
    plt.xlabel("Marketing Spend (in $1000s)")
    plt.ylabel("Sales Revenue (in $1000s)")
    plt.legend()
    plt.grid(True)
```

	plt.show() else: print("Correlation too weak. Regression not applied.")	
	Inferences: Q1: Is there a strong linear relationship between marketing spend and sales revenue? Q2: Can we predict sales revenue using a regression equation? Q3: How well does the regression model explain the variation in sales revenue?	
Solution:	Q1: You calculated the Pearson correlation coefficient: corr, _ = pearsonr(x, y) print("Pearson Correlation Coefficient:", round(corr, 3)) Let's say the output was: Pearson Correlation Coefficient: 0.936 Inference: Since r = 0.936 (which is > 0.7), there is a very strong positive linear relationship between marketing spend and sales revenue. This means as marketing spend increases, sales tend to increase proportionally. Q2: You used linregress to compute the regression line: slope, intercept, r_value, p_value, std_err = linregress(x, y) print(f"Regression Line: y = {slope:.2f}x + {intercept:.2f}") Sample output: Regression Line: y = 2.97x + 1.18 Inference: The regression equation can now be used to predict sales based on any given marketing spend. For example, if a company spends $60,000 on marketing: y=2.97×60+1.18=179.38 y=2.97×60+1.18=179.38 So, predicted sales = $179,380 Q3: How well does the regression model explain the variation in sales revenue?	

	You computed the coefficient of determination (R^2): print("R^2 =", round(r_value**2, 3)) Sample output: R^2 = 0.876 Inference: About 87.6% of the variability in sales is explained by the linear relationship with marketing spend. This indicates a very strong model fit, and suggests marketing budget is a strong predictor of sales. Overall Insights from the Case Study: Insight Explanation Strong correlation Pearson r = 0.936 confirms strong linear relationship Accurate prediction Regression equation enables useful forecasts Good model fit R^2 = 87.6% means the model captures most of the variation	